

# Absorbing Markov chains

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# Contents

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Fundamental matrix

Expected time to absorption

Absorption probabilities

Gambler's ruin

# Absorbing Markov chains

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## Definition

A finite Markov chain is called absorbing if:

- ▶ There exists at least an state  $j$  such that  $p_{jj} = 1$  (absorbing state).
  - ▶ From every state it is possible to reach an absorbing state (not necessarily in one step).
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- ▶ The absorbing states are exactly the **recurrent** states.
  - ▶ Once the chain enters an absorbing state  $j$ , i.e.  $X_n = j$ , their future evolution is totally determined:

$$X_{n+1} = j, X_{n+2} = j, X_{n+3} = j, \dots$$

## Canonical probability transition matrix

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If a Markov chain has  $r$  absorbing states and  $t$  transient states, we can arrange the transition matrix  $\mathbf{P}$  in a canonical form:

$$\mathbf{P} = \begin{pmatrix} \mathbf{Q} & \mathbf{R} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$$

where

- ▶  $\mathbf{Q}$  is a  $t \times t$  block containing the transition probabilities between transient states.
- ▶  $\mathbf{R}$  is a  $t \times r$  block containing the transition probabilities from transient to absorbing states.
- ▶  $\mathbf{I}$  is the identity matrix of order  $r$  and  $\mathbf{0}$  is a block with null elements.

# Canonical probability transition matrix

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## Theorem

We have that

$$\lim_{n \rightarrow \infty} \mathbf{Q}^n = \mathbf{0}.$$

*Furthermore, the chain will be absorbed with probability 1. This means that one of the absorbing states will be visited almost surely.*

► Notice that  $\lim_{n \rightarrow \infty} \mathbf{Q}^n = \mathbf{0}$  means

$$\lim_{n \rightarrow \infty} r_{ij}(n) = 0$$

for any pair  $i, j$  of transient states.

# Fundamental matrix

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## Theorem

*For an absorbing Markov chain, the matrix  $I - Q$  has an inverse  $N$ , and*

$$N = \sum_{n=0}^{\infty} Q^n$$

- ▶  $N = (I - Q)^{-1}$  is called the **fundamental matrix**.
- ▶ The convergence of the series is element-wise.

## Fundamental matrix

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### Proof:

Firstly, let us prove that  $I - Q$  has an inverse by proving that the homogeneous linear system

$$(I - Q)x = 0$$

has  $x = 0$  as a unique solution.

Indeed,  $(I - Q)x = 0$  is equivalent to  $x = Qx$ , and we have:

$$x = Qx = Q^2x = \dots = Q^n x = \dots$$

Since  $Q^n x \rightarrow 0$  because  $Q^n \rightarrow 0$ , we conclude that  $x = 0$ .

## Fundamental matrix

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Now consider

$$I - Q^{n+1} = (I - Q)(I + Q + Q^2 + \cdots + Q^n).$$

By multiplying both sides of the equation on the left by  $N = (I - Q)^{-1}$ , we obtain:

$$N - NQ^{n+1} = I + Q + Q^2 + \cdots + Q^n$$

Therefore, letting  $n \rightarrow \infty$ ,

$$N = I + Q + Q^2 + Q^3 + \cdots = \sum_{n=0}^{\infty} Q^n$$

## Expected time to absorption

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Let  $\mathcal{T}$  be the set of transient states, and let  $i, j \in \mathcal{T}$ .

### Theorem

- ▶ The entry  $n_{ij}$  of the matrix  $\mathbf{N}$  represents the expected number of times the Markov chain occupies state  $j$ , given that it starts in state  $i$ , including the initial state when  $i = j$ .
- ▶ Therefore, the expected number of steps required for the chain to reach an absorbing state is given by

$$\tau_i = \sum_{j \in \mathcal{T}} n_{ij}.$$

Notice that if  $\boldsymbol{\tau} = (\tau_1, \dots, \tau_t)^t$  and  $\mathbf{c} = (1, \dots, 1)^t$ , then  $\boldsymbol{\tau} = \mathbf{N} \mathbf{c}$ .

## Expected time to absorption

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Proof:

If  $I(X_n = j)$  is the indicator random variable of the event  $\{X_n = j\}$ , then  $T_j = \sum_{n=0}^{\infty} I_n$  is the total number of visits to  $j$ .

Therefore,

$$\begin{aligned}\mathbb{E}(T_j | X_0 = i) &= \mathbb{E}\left(\sum_{n=0}^{\infty} I(X_n = j) \mid X_0 = i\right) \\ &= \sum_{n=0}^{\infty} \mathbb{E}(I(X_n = j) | X_0 = i) \\ &= \sum_{n=0}^{\infty} \mathbb{P}(X_n = j | X_0 = i) = \sum_{n=0}^{\infty} r_{ij}(n) = n_{ij}\end{aligned}$$

(Notice that  $r_{ij}(0) = 1$  if  $i = j$  and  $r_{ij}(0) = 0$  otherwise.)

## Expected time to absorption

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Let  $\mathcal{A}$  denote the set of absorbing states, let

$$T = \min\{n \geq 1: X_n \in \mathcal{A}\}$$

and let

$$\tau_i = \mathbb{E}(T \mid X_0 = i)$$

Therefore, since  $T = \sum_{j \in \mathcal{T}} T_j$ , we have that

$$\tau_i = \mathbb{E}(T \mid X_0 = i) = \sum_{j \in \mathcal{T}} \mathbb{E}(T_j \mid X_0 = i) = \sum_{j \in \mathcal{T}} n_{ij}$$

# Absorption probabilities

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## Theorem

*Let  $b_{ij}$  denote the probability that an absorbing Markov chain is absorbed in the absorbing state  $j$  given that it starts in the transient state  $i$ .*

*If  $\mathbf{B}$  is the  $t \times r$  matrix whose entries are  $b_{ij}$ , then*

$$\mathbf{B} = \mathbf{NR},$$

*where  $\mathbf{N}$  is the fundamental matrix and  $\mathbf{R}$  is as in the canonical form of the transition matrix  $\mathbf{P}$ .*

## Absorption probabilities

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Proof:

By conditioning on the first step, we deduce that, for any  $i \in \mathcal{T}$  and any  $j \in \mathcal{A}$ ,

$$b_{ij} = p_{ij} + \sum_{k \in \mathcal{T}} p_{ik} b_{kj}$$

Thus,

$$\mathbf{B} = \mathbf{R} + \mathbf{QB}$$

That is,

$$(\mathbf{I} - \mathbf{Q})\mathbf{B} = \mathbf{R}$$

Therefore,

$$\mathbf{B} = (\mathbf{I} - \mathbf{Q})^{-1}\mathbf{R} = \mathbf{NR}$$

## Absorption probabilities

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### Example:

Consider a chain with state space  $S = \{0, 1, 2, 3, 4\}$  and transition probability matrix

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

- This Markov chain represents a symmetric random walk with absorbing barriers at states 0 and 4.

## Absorption probabilities

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Reordering  $S$  as  $\{1, 2, 3, 0, 4\}$ , the matrix  $\mathbf{P}$  is in canonical form:

$$\left( \begin{array}{ccc|cc} 0 & 1/2 & 0 & 1/2 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1/2 & 0 & 0 & 1/2 \\ \hline 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right)$$

Then,

$$\mathbf{Q} = \begin{pmatrix} 0 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 0 \end{pmatrix} \quad \mathbf{R} = \begin{pmatrix} 1/2 & 0 \\ 0 & 0 \\ 0 & 1/2 \end{pmatrix}$$

## Absorption probabilities

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The fundamental matrix is

$$\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1} = \begin{pmatrix} 3/2 & 1 & 1/2 \\ 1 & 2 & 1 \\ 1/2 & 1 & 3/2 \end{pmatrix}$$

For example,

- ▶  $n_{23} = 1$  is the expected number of visits to state 3, given that the chain starts at state 2.
- ▶  $n_{11} = 3/2$  is the expected number of times the chain is in state 1, given that it starts at this state (so,  $X_0 = 1$  counts).

## Absorption probabilities

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Moreover

$$\tau = \begin{pmatrix} 3/2 & 1 & 1/2 \\ 1 & 2 & 1 \\ 1/2 & 1 & 3/2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 3 \end{pmatrix}$$

is the column vector whose elements are the expected times  $\tau_i$ .

- For instance, given  $\{X_0 = 2\}$ , the expected time to absorption is  $\tau_2 = 4$ .

## Absorption probabilities

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Finally, from

$$\mathbf{B} = \mathbf{NR} = \begin{pmatrix} 3/2 & 1 & 1/2 \\ 1 & 2 & 1 \\ 1/2 & 1 & 3/2 \end{pmatrix} \begin{pmatrix} 1/2 & 0 \\ 0 & 0 \\ 0 & 1/2 \end{pmatrix} = \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \\ 1/4 & 3/4 \end{pmatrix}$$

we obtain the different probabilities of absorption.

- For example, the first row of  $\mathbf{B}$  informs us that, conditional on  $\{X_0 = 1\}$ , the probabilities that the chain is absorbed in states 0 and 4 are  $3/4$  and  $1/4$ , respectively.

## Gambler's ruin

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Consider an absorbing Markov chain such that

$$S = \{0, 1, \dots, m\}$$

and with transition probabilities

$$p_{ij} = \begin{cases} 1, & j = i = 0 \\ p, & j = i + 1, 1 \leq i \leq m - 1 \\ q = 1 - p, & j = i - 1, 1 \leq i \leq m - 1 \\ 1, & j = i = m \end{cases}$$

(We assume  $0 < p < 1$ .)

Let us calculate the probability  $v(j)$  of being absorbed into state  $m$ , given that we start from state  $j$ .

## Gambler's ruin

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Notice that  $v(0) = 0$  and  $v(m) = 1$ . Moreover, by a first-step analysis, we deduce that, for  $1 \leq j \leq m-1$ ,

$$v(j) = p v(j+1) + q v(j-1)$$

This difference equation can be reformulated as:

$$p(v(j+1) - v(j)) = q(v(j) - v(j-1))$$

Hence,

$$\begin{aligned} v(j+1) - v(j) &= \left(\frac{q}{p}\right) (v(j) - v(j-1)) \\ &= \dots = \left(\frac{q}{p}\right)^j (v(1) - v(0)) = \left(\frac{q}{p}\right)^j v(1) \end{aligned}$$

## Gambler's ruin

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Moreover,

$$\begin{aligned} 1 = v(m) - v(0) &= \sum_{j=0}^{m-1} (v(j+1) - v(j)) \\ &= v(1) \sum_{j=0}^{m-1} \left(\frac{q}{p}\right)^j = \begin{cases} v(1)m, & \text{if } p = q \\ v(1) \frac{1 - (q/p)^m}{1 - q/p}, & \text{if } p \neq q \end{cases} \end{aligned}$$

Thus,

$$v(1) = \begin{cases} \frac{1}{m}, & \text{if } p = q \\ \frac{1 - (q/p)}{1 - (q/p)^m}, & \text{if } p \neq q \end{cases}$$

## Gambler's ruin

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Therefore, for  $0 \leq j \leq m$ ,

$$\begin{aligned} v(j) &= \sum_{k=0}^{j-1} (v(k+1) - v(k)) = v(1) \sum_{k=0}^{j-1} \left(\frac{q}{p}\right)^k \\ &= \begin{cases} v(1)j = \frac{j}{m}, & \text{if } p = q \\ v(1) \frac{1 - (q/p)^j}{1 - q/p} = \frac{1 - (q/p)^j}{1 - (q/p)^m}, & \text{if } p \neq q \end{cases} \end{aligned}$$

## Gambler's ruin

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Of course, the probability of being absorbed at 0 is

$$1 - v(j) = \begin{cases} 1 - \frac{j}{m}, & p = q \\ \frac{(q/p)^j - (q/p)^m}{1 - (q/p)^m}, & p \neq q \end{cases}$$

## Gambler's ruin

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Finally, let us calculate the expected time to absorption in the symmetric case ( $p = q$ ).

Let  $T = \min\{n \geq 0: X_n = 0 \text{ or } X_n = m\}$ , and let

$$\tau(i) = \mathbb{E}(T \mid X_0 = i)$$

By a first-step analysis, we find that for  $1 \leq i \leq m - 1$ :

$$\tau(i) = \frac{1}{2}(\tau(i+1) + 1) + \frac{1}{2}(\tau(i-1) + 1)$$

Moreover,  $\tau(0) = \tau(m) = 0$ .

## Gambler's ruin

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If we define  $d(i) = \tau(i) - \tau(i - 1)$ , then

$$d(i + 1) = d(i) - 2, \quad 1 \leq i \leq m - 1$$

Hence,

$$d(2) = d(1) - 2$$

$$d(3) = d(2) - 2 = d(1) - 4$$

.....

$$d(i) = d(1) - 2(i - 1)$$

.....

$$d(m) = d(1) - 2(m - 1)$$

## Gambler's ruin

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Moreover,

$$\begin{aligned} 0 &= \tau(m) - \tau(0) = d(m) + \cdots + d(1) \\ &= m d(1) - 2 \sum_{j=1}^m (j-1) \\ &= m d(1) - 2 \frac{m(m-1)}{2} \end{aligned}$$

Then,

$$d(1) = m - 1$$

## Gambler's ruin

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Finally, we get:

$$\begin{aligned}\tau(i) &= d(1) + d(2) + \cdots + d(i) \\ &= (m-1) + (m-3) + \cdots + (m-(2i-1)) \\ &= \frac{((m-1) + (m-2i+1)) i}{2} \\ &= i(m-i)\end{aligned}$$

For example, if  $m = 4$  we have  $\tau(1) = 3$ ,  $\tau(2) = 4$ ,  $\tau(3) = 3$ , as calculated in a previous example.